

Project Coversheet

Full Name	Modinat Omolara Buhari
Project Title (Example – Week1, Week2, Week3, Week 4)	Week 2: Sales & Customer Behaviour Insights – Green Cart Ltd.

Instructions:

Students must download this cover sheet, use it as the first page of their project, and then save the entire document as a PDF before submission.

Project Guidelines and Rules

1. Formatting and Submission

Format: Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.

Title: Include Week and Title (Example - Week 1: Travel Ease Case Study.)

File Format: Submit as PDF or Word file

Page Limit: 4–5 pages, including the title and references.

2. Answer Requirements

Word Count: Each answer should be within 100–150 words; Maximum 800–1,200 words.

Clarity: Write concise, structured answers with key points.

Tone: Use formal, professional language.

3. Content Rules

Answer all questions thoroughly, referencing case study concepts.

Use examples where possible (e.g., risk assessment techniques).

Break complex answers into bullet points or lists.

4. Plagiarism Policy

Submit original work; no copy-pasting.

Cite external material in a consistent format (e.g., APA, MLA).

5. Evaluation Criteria

Understanding: Clear grasp of business analysis principles.

Application: Effective use of concepts like cost-benefit analysis and Agile/Waterfall.

Clarity: Logical, well-structured responses.

Creativity: Innovative problem-solving and examples.

Completeness: Answer all questions within the word limit.

6. Deadlines and Late Submissions

Deadline: Submit on time; trainees who fail to submit the project will miss the "Certificate of Excellence"

7. Additional Resources

Refer to lecture notes and recommended readings.

Contact the instructor or peers for clarifications before the deadline.

1. Introduction

Green Cart Ltd. is a UK-based e-commerce company that sells eco-friendly household products. The objective of this project was to analyze sales performance, customer purchasing behaviour, and delivery performance using sales, product, and customer datasets.

The project involved cleaning and merging datasets, creating new analytical features, generating summary tables, and producing visualizations to identify trends and business opportunities. The analysis focused on revenue performance, customer loyalty, delivery efficiency, and product category performance. The findings provide insights that can support future marketing strategies, inventory planning, and customer retention initiatives.

2. Data Cleaning Summary

Data cleaning was performed on all three datasets before analysis. Text fields such as delivery status, payment methods, regions, and loyalty tiers were standardized to ensure consistency. Date columns were converted into datetime format for accurate calculations and trend analysis.

Missing values were identified and handled appropriately. Discount values were replaced with 0 where missing, while missing categorical values were filled with "Unknown" when necessary. Duplicate records were removed from the sales and customer datasets. Numeric columns, including quantity, unit price, and discount, were validated to ensure they contained no negative values.

The cleaning process improved data quality and ensured reliable results throughout the analysis.

3. Feature Engineering Summary

Several new features were created to support deeper business analysis:

Revenue: Calculated as $\text{Quantity} \times \text{Unit Price} \times (1 - \text{Discount Applied})$.

Order Week: Extracted from the order date to support weekly trend analysis.

Price Band: Products categorized into Low, Medium, and High price groups.

Days to Order: Number of days between product launch and purchase date.

Email Domain: Extracted from customer email addresses.

Is Late: Boolean flag identifying delayed deliveries.

These engineered features enabled more meaningful insights into customer behaviour, product performance, and operational efficiency.

4. Key Findings & Trends

The analysis revealed several important trends. The Cleaning category generated the highest revenue (£93,621.78), significantly outperforming all other categories. Storage and Outdoors products were the second and third highest revenue generators.

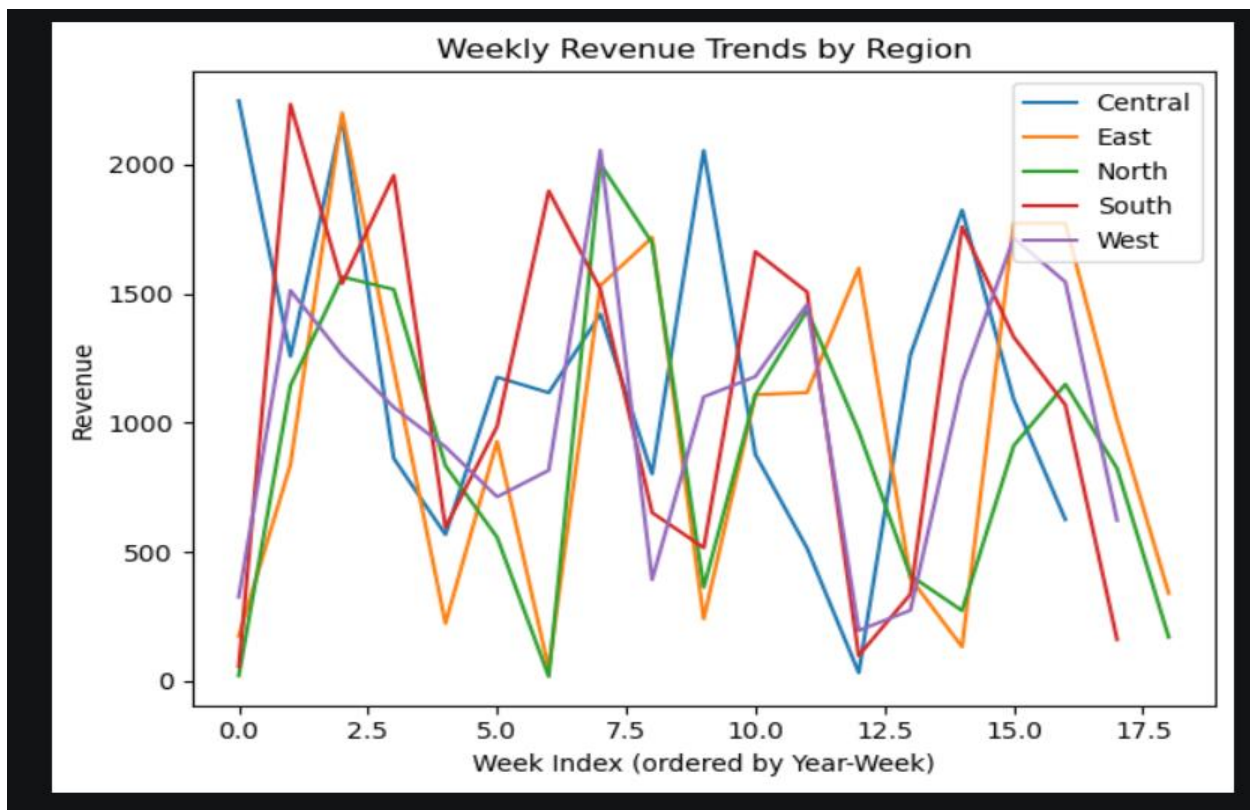
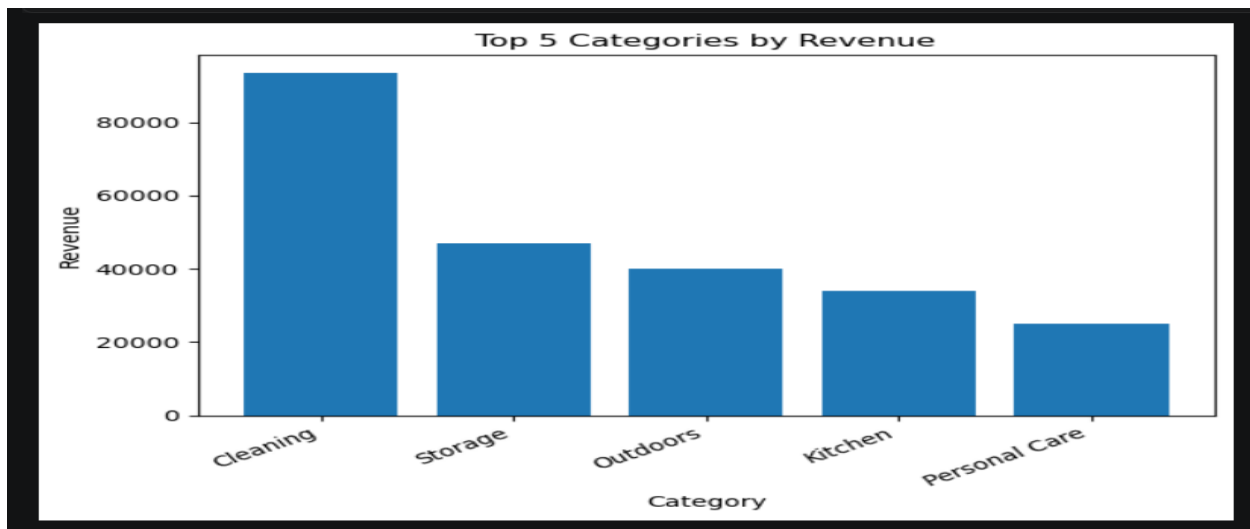
Weekly revenue varied considerably across regions, indicating seasonal and regional purchasing patterns. Some weeks experienced strong revenue growth, suggesting opportunities for targeted promotions during peak periods.

Delivery analysis showed that delayed deliveries occurred across all price bands, with High and Medium-priced products experiencing the largest number of delays. This suggests potential operational challenges that may impact customer satisfaction and retention.

5. Business Question Answers

a) Which product categories drive the most revenue and in which regions?

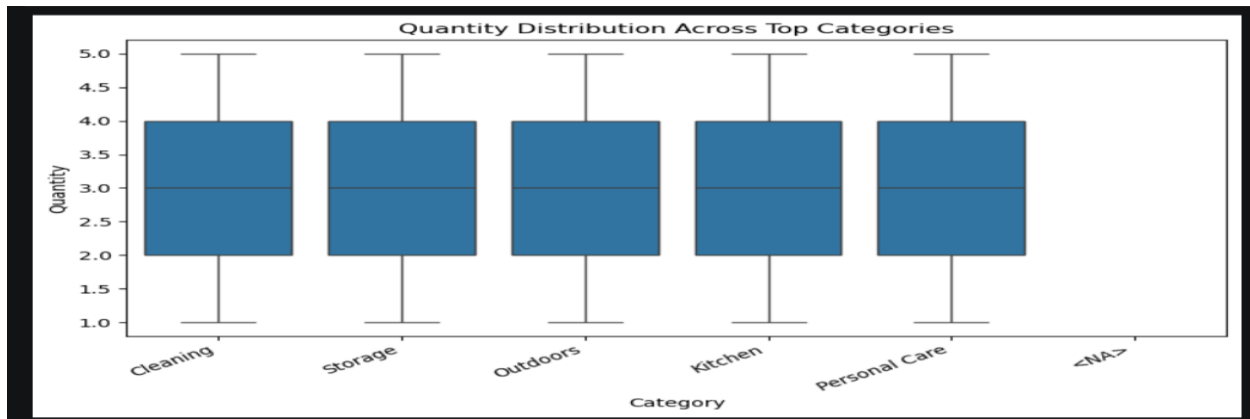
The Cleaning category generated the highest revenue (£93,621.78), followed by Storage (£46,931.46) and Outdoors (£40,062.07). Revenue was distributed across multiple regions, with strong contributions from Central, South, and West regions. Cleaning products appear to be the company's strongest-performing product line and should remain a strategic focus.



b) Do discounts lead to more items sold?

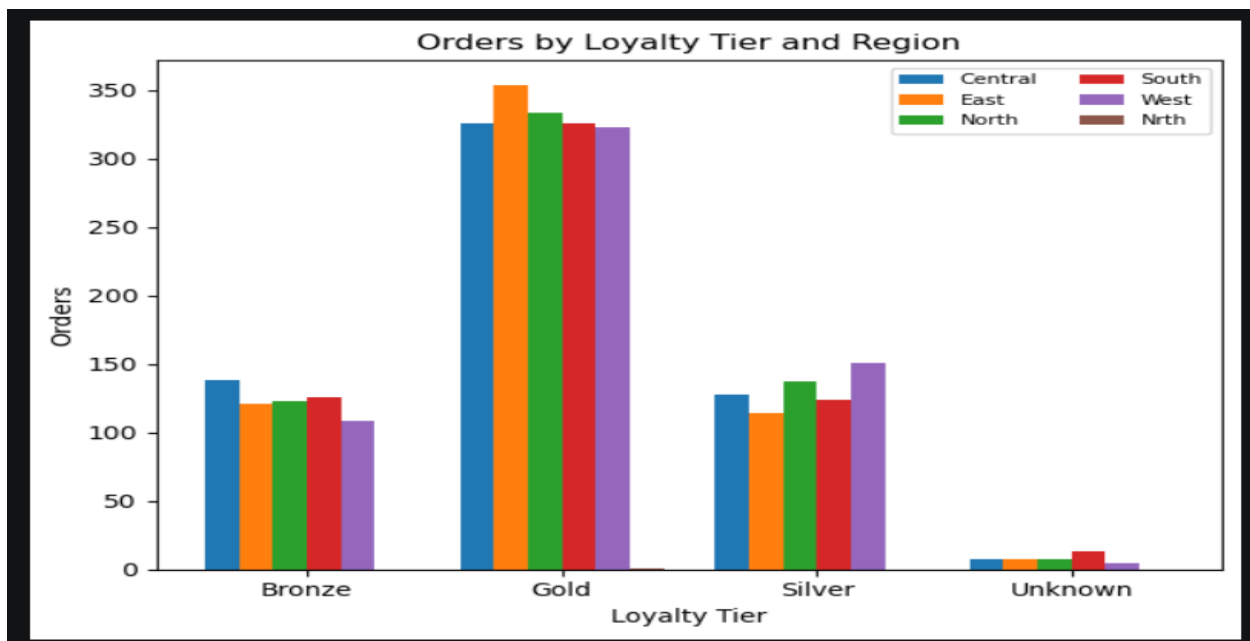
The relationship between quantity sold and discounts suggests that discounted products generally achieved higher sales volumes. While discounts can reduce profit margins, they appear to encourage customer purchases and increase product movement.

Strategic discount campaigns may therefore support sales growth when applied carefully.



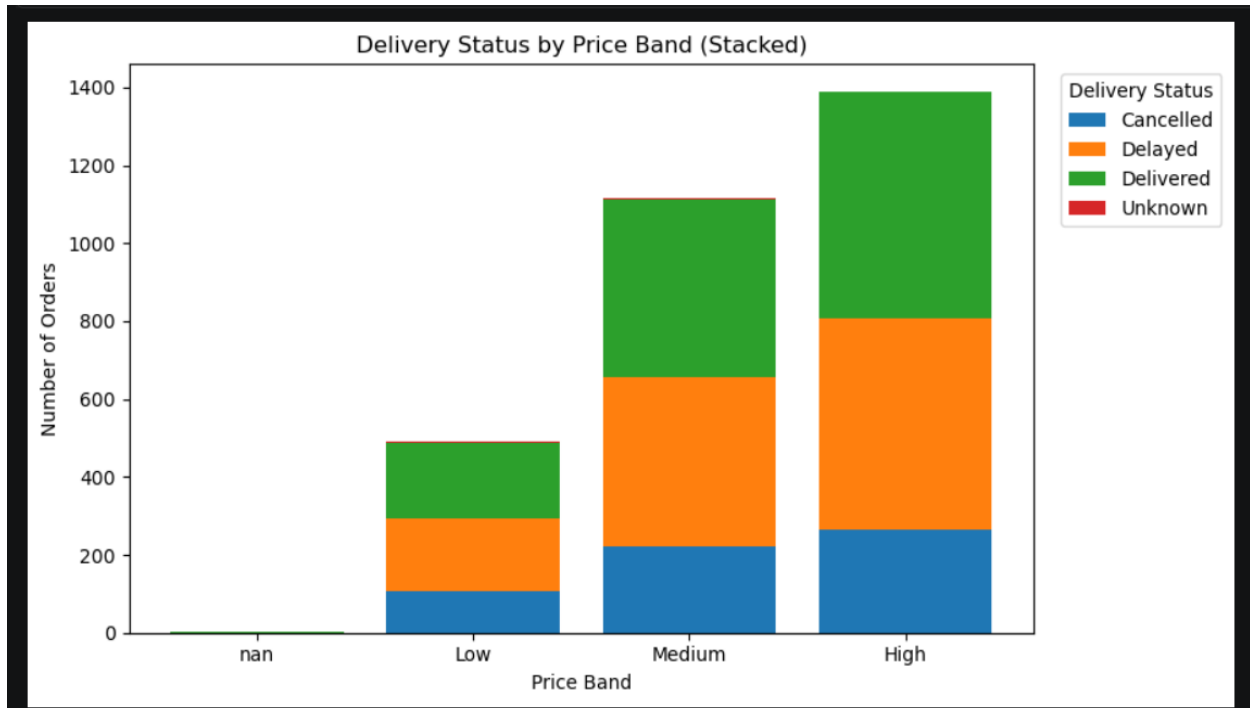
c) Which loyalty tier generates the most value?

Analysis of customer behaviour by loyalty tier indicates that higher loyalty tiers contributed more revenue and order activity. Customers in premium loyalty segments tend to make more purchases and generate greater business value. Strengthening loyalty programs could improve customer retention and long-term revenue.



d) Are certain regions struggling with delivery delays?

Yes. Delivery performance analysis showed delays occurring in several regions. Medium and High-priced product categories experienced particularly high numbers of delayed deliveries. Regions with frequent delays should be reviewed to identify logistics bottlenecks and improve customer experience.



6. Recommendations

Increase focus on the Cleaning product category through targeted marketing campaigns and inventory optimization, as it is the strongest revenue driver.

Expand loyalty program benefits to encourage customer retention and increase spending among high-value customers.

Investigate delivery delays in regions with poor delivery performance and implement process improvements to improve customer satisfaction.

Use strategic discounts on selected products to increase sales volume while monitoring profitability.

7. Data Issues or Risks

One significant data quality issue was inconsistent formatting and missing values across multiple datasets. Examples included inconsistent delivery status labels, loyalty tier formatting, missing discount values, and incomplete customer information.

To reduce data quality issues, Green Cart Ltd. should implement stronger validation rules during data entry, standardize dropdown selections for categorical fields, and introduce automated quality checks before data is stored in the company database.

Conclusion

This analysis provided valuable insights into Green Cart Ltd.'s sales performance, customer behaviour, and delivery operations. The Cleaning category emerged as the strongest revenue contributor, loyalty customers generated significant business value, and delivery delays were identified as an operational concern. By acting on these findings, Green Cart Ltd. can improve profitability, customer satisfaction, and overall business performance.

References

1. Green Cart Ltd. (2026). Sales and Customer Behaviour Insights Dataset (sales_data.csv, product_info.csv, customer_info.csv). Internal company dataset provided for analysis.
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